PTF ALL

Equip Cooling Supply Fan Does Not Shut Off During A Cargo Compartment Smoke Alert - Fault Isolation

CORRECTIVE ACTION SUMMARY

Replace Supply System Control Interrupt Relay [R645]
Replace Aft Fire Alarm Latch Relay [R946]
Replace Forward Fire Alarm Latch Relay [R945]
Replace Recirculate Fan Switch Bypass Relay [R644]
Replace E6 Rack Fan Interrupt Relay [R1205]
Replace E6 Rack Fan Command Invert Relay [R1206]
Replace Smoke Control Relay [R648]
Replace Isolation Diode [R958]
Repair Wiring D11868-D12227
Repair Wiring D12227-D12225
Repair Wiring D12225-D11866
Repair Wiring D11866-D11874
Repair Wiring D11874-Circuit Breaker C1444
Repair Wiring Circuit Breaker C4070-D14139
Repair Wiring D14139-D14141
Repair Wiring for D14141

A. Initial Evaluation

(1) Do this task:
Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807.

(a) If the system does not operate as specified in the AMM task, then do the Fault Isolation Procedure below.

(b) If the system operates as specified in the AMM task, then there was an intermittent fault.

B. Fault Isolation Procedure

(1) Replace the supply system control interrupt relay, R645.
(a) Do the AMM test that you did at the start of this task.
(b) If the system operates as specified in the AMM task, then you corrected the fault.
(c) If the system does not operate as specified in the AMM task, then continue.

(2) Replace the aft fire alarm latch relay, R946.
(a) Do the AMM test that you did at the start of this task.
(b) If the system operates as specified in the AMM task, then you corrected the fault.
(c) If the system does not operate as specified in the AMM task, then continue.

(3) Replace the forward fire alarm latch relay, R945.
(a) Do the AMM test that you did at the start of this task.
(b) If the system operates as specified in the AMM task, then you corrected the fault.
(c) If the system does not operate as specified in the AMM task, then continue.

(4) Replace the recirculate fan switch bypass relay, R644.
(a) Do the AMM test that you did at the start of this task.
(b) If the system operates as specified in the AMM task, then you corrected the fault.
(c) If the system does not operate as specified in the AMM task, then continue.

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(5) Replace the E6 rack fan interrupt relay, R1205.
   (a) Do the AMM test that you did at the start of this task.
   (b) If the system operates as specified in the AMM task, then you corrected the fault.
   (c) If the system does not operate as specified in the AMM task, then continue.

(6) Replace the E6 rack fan command invert relay, R1206.
   (a) Do the AMM test that you did at the start of this task.
   (b) If the system operates as specified in the AMM task, then you corrected the fault.
   (c) If the system does not operate as specified in the AMM task, then continue.

(7) Replace the smoke control relay, R648.
   (a) Do the AMM test that you did at the start of this task.
   (b) If the system operates as specified in the AMM task, then you corrected the fault.
   (c) If the system does not operate as specified in the AMM task, then continue.

(8) Do this check of the isolation diode:
   (a) Disconnect the wires from the power terminals of the isolation diode, R958 (WDM 21-43-21).
   (b) Connect an ohmmeter to do this diode check in the forward direction:
      1) Connect the positive probe of the meter to terminal A19 of the isolation diode.
      2) Connect the negative probe of the meter to terminal B19 of the isolation diode.
      3) There will be continuity between the pins if the forward direction is good.
   (c) Connect the ohmmeter to do this diode check in the reverse direction:
      1) Connect the negative probe of the meter to terminal A19 of the isolation diode.
      2) Connect the positive probe of the meter to terminal B19 of the isolation diode.
      3) There will be no continuity between the pins if the reverse direction is good.
   (d) If the diode check was not good in the forward direction or the reverse direction, then do these steps:
      1) Replace the isolation diode, R958.
      2) Do the AMM test that you did at the start of this task.
      3) If the system operates as specified in the AMM task, then you corrected the fault.
   (e) If the diode check was good in the forward and the reverse directions, then continue.
   (f) Re-connect the wires to the power terminals of the isolation diode, R958.

(9) Do this check of the wiring:
   (a) Disconnect connector D11868 from the supply system control interrupt relay, R645.
   (b) Disconnect connector D12227 from the aft fire alarm latch relay, R946.
   (c) Do a wiring check between these pins of connector D11868 at the supply system control interrupt relay and these pins of connector D12227 at the aft fire alarm latch relay (WDM 21-27-31, WDM 21-43-21):

D11868		**D12227**
pin X1	.....	pin 14
pin X1	.....	pin 9

   (d) If you find a problem with the wiring, then do these steps:
      1) Repair the wiring.
      2) Re-connect connector D11868 to the supply system control interrupt relay, R645.
      3) Re-connect connector D12227 to the aft fire alarm latch relay, R946.
      4) Do the AMM test that you did at the start of this task.
      5) If the system operates as specified in the AMM task, then you corrected the fault.
   (e) If you do not find a problem with the wiring, then continue.
   (f) Re-connect connector D11868 to the supply system control interrupt relay, R645.

(10) Do this check of the wiring:
   (a) Disconnect connector D12225 from the forward fire alarm latch relay, R945.

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(b) Do a wiring check between these pins of connector D12227 at the aft fire alarm latch relay and these pins of connector D12225 at the forward fire alarm latch relay (WDM 21-27-31, WDM 21-43-21):

D12227		D12225
pin 7	.....	pin 14
pin 11	.....	pin 9

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D12227 to the aft fire alarm latch relay, R946.
3) Re-connect connector D12225 to the forward fire alarm latch relay, R945.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D12227 to the aft fire alarm latch relay, R946.

(11) Do this check of the wiring:

(a) Disconnect connector D11866 from the recirculate fan switch bypass relay, R644.

(b) Do a wiring check between these pins of connector D12225 at the forward fire alarm latch relay and these pins of connector D11866 at the recirculate fan switch bypass relay (WDM 21-43-21):

D12225		D11866
pin 7	.....	pin 4
pin 11	.....	pin 14

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D12225 to the forward fire alarm latch relay, R945.
3) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D12225 to the forward fire alarm latch relay, R945.

(12) Do this check of the wiring:

(a) Disconnect connector D11874 from the smoke control relay, R648.

(b) Do a wiring check between these pins of connector D11866 at the recirculate fan switch bypass relay and these pins of connector D11874 at the smoke control relay (WDM 21-43-21):

D11866		D11874
pin 5	.....	pin B1
pin 7	.....	pin C1

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.
3) Re-connect connector D11874 to the smoke control relay, R648.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.

(13) Do this check of the wiring:

(a) Open the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].

(b) Do a wiring check between these pins of connector D11874 at the smoke control relay and this terminal of circuit breaker C1444 (WDM 21-43-21):

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D11874		C1444
pin B2	.....	term =L
pin C2	.....	term =L

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D11874 to the smoke control relay, R648.
3) Close the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D11874 to the smoke control relay, R648.

(f) Close the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].

(14) Do this check of the wiring:

(a) Open the EQUIPMENT COOLING E6 RACK CONTROL circuit breaker, C4070 [P6-5, D10].

(b) Disconnect connector D14139 from the E6 rack fan interrupt relay, R1205.

(c) Do a wiring check between this terminal of circuit breaker C4070 and this pin of connector D14139 at the E6 rack fan interrupt relay (WDM 21-27-51):

C4070		D14139
term =L	.....	pin 6

(d) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Close the EQUIPMENT COOLING E6 RACK CONTROL circuit breaker, C4070 [P6-5, D10].
3) Re-connect connector D14139 to the E6 rack fan interrupt relay, R1205.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

(f) Close the EQUIPMENT COOLING E6 RACK CONTROL circuit breaker, C4070 [P6-5, D10].

(15) Do this check of the wiring:

(a) Disconnect connector D14141 from the E6 rack fan command invert relay, R1206.

(b) Do a wiring check between this pin of connector D14139 at the E6 rack fan interrupt relay and this pin of connector D14141 at the E6 rack fan command invert relay (WDM 21-27-51):

D14139		D14141
pin 12	.....	pin 5

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D14139 to the E6 rack fan interrupt relay, R1205.
3) Re-connect connector D14141 to the E6 rack fan command invert relay, R1206.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D14139 to the E6 rack fan interrupt relay, R1205.

(16) Do this check of the wiring:

(a) Disconnect connector D12227 from the aft fire alarm latch relay, R946.

(b) Do a wiring check between this pin of connector D14141 at the E6 rack fan command invert relay and this pin of connector D12227 at the aft fire alarm latch relay (WDM 21-27-51, WDM 21-43-21).
1) Connect the positive probe of the meter to this pin of connector D12227 at the aft fire alarm latch relay.

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2) Connect the negative probe of the meter to this pin of connector D14141 at the E6 rack fan command invert relay.

D14141		D12227
pin 7	.....	pin 8

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D14141 to the E6 rack fan command invert relay, R1206.
3) Re-connect connector D12227 to the aft fire alarm latch relay, R946.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D12227 to the aft fire alarm latch relay, R946.

(17) Do this check of the wiring:

(a) Disconnect connector D12225 from the forward fire alarm latch relay, R945.

(b) Do a wiring check between this pin of connector D14141 at the E6 rack fan command invert relay and this pin of connector D12225 at the forward fire alarm latch relay (WDM 21-27-51, WDM 21-43-21).
1) Connect the positive probe of the meter to this pin of connector D12225 at the forward fire alarm latch relay.
2) Connect the negative probe of the meter to this pin of connector D14141 at the E6 rack fan command invert relay.

D14141		D12225
pin 7	.....	pin 8

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D14141 to the E6 rack fan command invert relay, R1206.
3) Re-connect connector D12225 to the forward fire alarm latch relay, R945.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

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